Cambridge (CIE) IGCSE Maths: Extended

Your notes

Number Toolkit

Contents

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Types of Numbers

Your notes

Types of Number

You will come across vocabulary such as

- **Integers** and **natural** numbers
- **Rational** and **irrational** numbers
- **Multiples**
- **Factors**
- **Prime** numbers
- **Squares**, **cubes** and **roots**
- **Reciprocals**

Knowing what each of these terms mean is essential.

What are integers and natural numbers?

- **Integers** are **whole** numbers;
  - They can be **positive, negative** and **zero**
  - For example, -3, -2, -1, 0, 1, 2, 3 are all integers
- **Natural** numbers are the **positive integers**
  - They can be thought of as counting numbers
  - 1, 2, 3, 4, ... are the natural numbers
    - Notice that 0 is **not** included

What are multiples?

- A **multiple** is a number which can be divided by another number, without leaving a remainder
  - For example, 12 is a multiple of 3
    - 12 divided by 3 is exactly 4
- A **common multiple** is multiple that is shared by more than one number
  - For example, 12 is a common multiple of 4 and 6

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

- **Even** numbers (2, 4, 6, 8, 10, ...) are multiples of 2
- **Odd** numbers (1, 3, 5, 7, 9, ...) are **not** multiples of 2
- Multiples can be **algebraic**
  - For example, the multiples of k would be $k, 2k, 3k, 4k, 5k. ...$

What are factors?

- A **factor** of a given number is a value that divides the given number exactly, with **no remainder**
  - 6 is a factor of 18
    - because 18 divided by 6 is exactly 3
- Every integer greater than 1 has at least two factors
  - The integer itself, and 1
- A **common factor** is a factor that is shared by more than one number
  - For example, 3 is a common factor of both 21 and 18

How do I find factors?

- Finding all the factors of a particular value can be done by finding **factor pairs**
- For example when finding the factors of 18
  - 1 and 18 will be the first factor pair
  - Divide by 2, 3, 4 and so on to test if they are factors
    - 18 ÷ 2 = 9, so 9 and 2 are factors
    - 18 ÷ 3 = 6, so 6 and 3 are factors
    - 18 ÷ 4 = 4.5, so 4 is not a factor
    - 18 ÷ 5 = 3.6, so 5 is not a factor
    - 18 ÷ 6 would be next, but we have already found that 6 was a factor
    - So we have now found all the factors of 18: 1, 2, 3, 6, 9

How do I find factors without a calculator?

- Use a **divisibility test**
  - Some tests are easier to remember, and more useful, than others

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

- Once you know that the number has a particular factor, you can divide by that factor to find the factor pair
- Instead of a divisibility test, you could use a formal written method to divide by a value
  - If the result is an integer; you have found a factor

What are some useful divisibility tests?

- A number is **divisible by 2** if the last digit is even (a multiple of 2)
- A number is **divisible by 3** if the sum of the digits is divisible by 3 (a multiple of 3)
  - 123
    $1+2+3=6$; 6 is a multiple of 3, so 123 is divisible by 3
  - 134
    $1+3+4=8$; 8 is not a multiple of 3, so 134 is not divisible by 3
- A number is **divisible by 4** if halving the number **twice** results in an integer
- A number is **divisible by 8** if it can be halved **3 times** and the result is an integer
- A number is **divisible by 5** if the last digit is a 0 or 5
- A number is **divisible by 10** if the last digit is a 0

What are prime numbers?

- A **prime** number is a number which has exactly two (**distinct**) **factors;** itself and 1
  - You should remember at least the first ten prime numbers:
    - 2, 3, 5, 7, 11, 13, 17, 19, 23, 29
- 1 is **not** a prime number, because:
  - by definition, prime numbers are integers greater than or equal to 2
  - 1 only has one factor
- 2 is the only even prime number
- If a number has any factors other than itself and 1, it is not a prime number

Your notes

Worked Example

Show that 51 is **not** a prime number.

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

If we can find a factor of 51 (that is not 1 or 51), this will prove it is not prime

51 is not even so is not divisible by 2
Next use the divisibility test for 3

$$5 + 1 = 6;\ 6 \text{ is divisible by 3; therefore 51 is divisible by 3}$$
$$51 \div 3 = 17$$

The factors of 51 are 1, 3, 17 and 51

51 is not prime as it has more than two (distinct) factors

What are square numbers?

- A **square** number is the result of multiplying a number by itself
  - The first square number is $1 \times 1 = 1$, the second is $2 \times 2 = 4$ and so on
- The first 15 square numbers are: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225
  - Aim to remember at least the first fifteen square numbers
- In algebra, square numbers can be written using a power of 2
  - $a \times a = a^2$

What are cube numbers?

- A **cube** number is the result of multiplying a number by itself, twice
  - The first cube number is $1 \times 1 \times 1 = 1$, the second is $2 \times 2 \times 2 = 8$ and so on
- The first 5 cube numbers are 1, 8, 27, 64 and 125
  - Aim to remember at least the first five cube numbers
  - You should also remember $10^3 = 1000$
- In algebra, cube numbers can be written using a power of 3
  - $a \times a \times a = a^3$

What are square roots?

- The **square root** of a value, is the number that when multiplied by itself equals that value
  - For example, 4 is the square root of 16
  - It is the opposite of squaring

Your notes

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

- Square roots are indicated by the symbol $\sqrt{}$
  - e.g. The square root of 49 would be written as $\sqrt{49}$
- Square roots can be **positive and negative**
  - e.g. The square roots of 25 are 5 and -5
- If a negative square root is required then a - sign would be used
  - e.g. $\sqrt{25} = 5$ but $-\sqrt{25} = -5$
  - Sometimes both positive and negative square roots are of interest and would be indicated by $\pm\sqrt{25}$
- The square root of a non-square **integer** is also called a **surd**
  - e.g. $\sqrt{3}$ is a surd, as 3 is not a square number
  - Surds are **irrational** numbers
  - $\sqrt{64}$ is **rational,** as it is equal to 8
  - However, $\sqrt{2}$ is **irrational**, as 2 is not a square number
- You should aim to remember the square roots of the first 15 square numbers
  - $\sqrt{1}, \sqrt{4}, \sqrt{9}, \sqrt{16}, \sqrt{25}, \sqrt{36}, \sqrt{49}, \sqrt{64}, \sqrt{81}, \sqrt{100}, \sqrt{121}, \sqrt{144}, \sqrt{169}, \sqrt{196}, \sqrt{225}$

What are cube roots?

- The **cube root** of a value, is the number that when multiplied by itself twice equals that value
  - For example, 3 is the cube root of 27
  - It is the opposite of cubing
  - Cube roots are indicated by the symbol $\sqrt[3]{}$
    - e.g. The cube root of 64 would be written as $\sqrt[3]{64}$
  - You should remember the values of the following cube roots:
    - $\sqrt[3]{1}, \sqrt[3]{8}, \sqrt[3]{27}, \sqrt[3]{64}, \sqrt[3]{125}, \sqrt[3]{1000}$

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

Worked Example

Write down a number which is both a cube number and a square number, and hence express this number in two different ways using index notation.

Listing the first 12 square numbers

1, 4, 9, 16, 25, 36, 49, **64**, 81, 100, 121, 144

Listing the first 5 cube numbers

1, 8, 27, **64**, 125

64 appears in both lists, it is the 8th square number and 4th cube number

64 is both a square and cube number
$64 = 8^2$ and $64 = 4^3$

What is a reciprocal?

- The **reciprocal** of a number is the result of **dividing 1 by that number**
    - Any number multiplied by its reciprocal will be equal to 1
        - The reciprocal of an integer is $\frac{1}{\text{integer}}$
        - The reciprocal of the fraction $\frac{a}{b}$ is $\frac{b}{a}$
        - The fraction is flipped upside-down!
- The reciprocal of 3 is $\frac{1}{3}$
    - The reciprocal of $\frac{1}{3}$ is 3
    - $3 \times \frac{1}{3} = \frac{1}{3} \times 3 = 1$
- The reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

- The reciprocal of $\frac{3}{2}$ is $\frac{2}{3}$
- $\frac{2}{3} \times \frac{3}{2} = \frac{3}{2} \times \frac{2}{3} = 1$
- Algebraically the reciprocal of a is $\frac{1}{a}$
  - The reciprocal of $\frac{1}{a}$ is a
  - This can also be written using a power of -1
    - $\frac{1}{a} = a^{-1}$

Worked Example

Write down a fraction that completes this calculation: $\frac{3}{7} \times \frac{\dots}{\dots} = 1$

Recall that a number multiplied by its reciprocal is equal to 1

$$\frac{3}{7} \times \mathbf{\frac{7}{3}} = 1$$

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

Irrational Numbers

What is a rational number?

- A **rational** number is a number that can be written as a fraction in its simplest form
  - It must be possible to write in the form $\frac{a}{b}$, where a and b are both **integers**
    - b cannot be zero
  - This includes all terminating and recurring decimals
    - E.g. 0.15 (which is $\frac{15}{100}$) and 0.151515151515... (which is $\frac{15}{99}$)
  - This also includes all integers
    - 5 can be written as $\frac{5}{1}$
    - -3 can be written as $\frac{-3}{1}$ or $\frac{3}{-1}$
    - 0 can be written as $\frac{0}{1}$ (it is ok to have 0 in the numerator)

What is an irrational number?

- An **irrational** number is a number that **cannot** be written in the form $\frac{a}{b}$, where a and b are **integers** and $\frac{a}{b}$ is in its simplest form
  - A decimal which is non-terminating and non-recurring is an irrational number
  - The number $\sqrt{n}$, where n is not a square number, is an irrational number
    - This is also known as a **surd**

© 2015-2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

What irrational numbers should I know?

Your notes

- You may be asked to identify an irrational number from a list
- Irrational numbers that you should recognise are π, $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$,
  - Any multiple of these is also irrational
    - For example 2π, $3\sqrt{2}$, $3\sqrt{5}$ are all irrational
  - Be careful: $\sqrt{2} \times \sqrt{2}$ is not irrational as it equals 2
- Most calculators will show irrational numbers in their exact form rather than as a decimal

Examiner Tips and Tricks

If you're not sure if a number is rational or irrational, type it into your calculator and see if it can be displayed as a fraction.

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

Negative Numbers

What are the rules for calculations with negative numbers?

- When **multiplying** and **dividing** with negative numbers
  - Two numbers with the **same sign** make a **positive**
    - $(-12) \div (-4) = 3$
    - $(-6) \times (-4) = 24$
  - Two numbers with **different signs** make a **negative**
    - $(-12) \div 4 = -3$
    - $6 \times (-4) = -24$
- When **adding** and **subtracting** with negative numbers
  - **Subtracting a negative** number is the same as **adding** the positive
    - e.g. $5 - (-3) = 5 + 3 = 8$
  - **Adding a negative** number is the same as **subtracting** the positive
    - e.g. $7 + (-3) = 7 - 3 = 4$

Where are negative numbers used in real-life?

- **Temperature** is a common context for negative numbers
  - If the temperature is 3°C, and it cools by 5°C, the new temperature will be -2°C
    - This is equivalent to 3 - 5 = - 2
  - If the temperature is -4°C, and it warms up by 6°C, the new temperature will be 2°C
    - This is equivalent to (-4) + 6 = 2
  - To explain why (-5) - (-6) = 1, you could think of it as follows:
    - A room is -5°C, then -6°C of cold air is 'removed'
    - The room now warms to 1°C

© 2015-2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

- **Money** and **debt** is another common context for negative numbers
  - A negative sign means you owe money
  - If someone has a debt of $200, and they borrow another $400, their total debt is now $600
    - This is equivalent to (-200) + (-400) = -600
  - If someone is in debt by $300, but then pays off $200 of their debt, they are now only $100 in debt
    - This is equivalent to (-300) + 200 = -100

Your notes

Examiner Tips and Tricks

- Your calculator isn't always as clever as you may think!
- Using brackets around negative numbers will always make sure the calculator is doing what you want
  - e.g. The square of negative three is $(-3) \times (-3) = 9$
    On many calculators, $-3^2 = -9$ but $(-3)^2 = 9$
    The second one is correct

Worked Example

Complete the following table.

Calculation	Answer
3 + (-4)	
(-5) + (-8)	
7 - (-10)	
(-8) - (-6)	
(-3) × 6	
(-9) × (-2)	
(-9) ÷ (-3)	
(-10) ÷ 5	

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Calculation	Working	Answer
3 + (-4)	3 - 4	**-1**
(-5) + (-8)	(-5) - 8	**-13**
7 - (-10)	7 + 10	**17**
(-8) - (-6)	(-8) + 6	**-2**
(-3) × 6	3 × 6 = 18 one is negative	**-18**
(-9) × (-2)	9 × 2 = 18 both are negative	**18**
(-9) ÷ (-3)	9 ÷ 3 = 3 both are negative	**3**
(-10) ÷ 5	10 ÷ 5 = 2 one is negative	**-2**

Your notes

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Mathematical Symbols

Your notes

Mathematical Symbols

What mathematical symbols do I need to know?

- Starting with the basic four operations
  - $+$ **Addition,** plus, sum, total
  - $-$ **Subtraction,** minus, difference, take away
  - $\times$ **Multiplication,** product, times
  - $\div$ **Division,** quotient, divide, share
- **Equals** signs
  - $=$ Equal to
    - e.g. $3x+7=19$
  - $\neq$ Not equal to
    - e.g. $2-5\neq 5-2$
  - $\approx$ Approximately equal to
    - e.g. $\pi\approx 3.14$
  - $\equiv$ Identical (equivalent) to
    - e.g. $12x+6\equiv 3(4x+2)$
- **Inequality** signs
  - $>$ **Greater** than
    - e.g. $5>-2$
  - $<$ **Less** than
    - e.g. $\frac{1}{2}<\frac{2}{3}$
  - $\geq$ Greater than or equal to

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

  - $\leq$ Less than or equal to
- Other helpful symbols
  - $(\ )$ **Brackets**
    - used to group symbols
    - e.g. $(2x + 4) - 3(x + 7)$
  - 3^x **Powers** (also called indices, orders, or exponents)
    - repeated multiplication
    - e.g. $3^4 = 3 \times 3 \times 3 \times 3 = 81$
  - $\sqrt{\ }$ **Square root**
    - opposite of squaring
    - e.g. $\sqrt{81} = 9$ is the opposite to $9^2 = 81$
  - $\pm$ **Plus Minus**
    - used to allow for both a positive and negative answer (two distinct answers)
    - $a \pm b$ is the same as $a + b$ and $a - b$
    - e.g. $x = 3 \pm 2.5$ is the same as $x = 3 + 2.5 = 5.5$ and $x = 3 = 2.5 = 0.5$
  - π **Pi**
    - the ratio of the circumference of a circle to its diameter

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

Order of Operations (BIDMAS/BODMAS)

What is the order of operations (BODMAS/BIDMAS)?

- If there is more than one **operation** in a calculation then they should be done in the following order
  - **B**rackets: ( )
    - Perform any calculation(s) **inside** brackets first
  - p**O**wers or **I**ndices (sometimes **O**rder): ^{2}, ^{3}, √ and similar
    - These include **powers, roots, reciprocals**
  - **D**ivisions or **M**ultiplications: × or ÷
    - If there are more than one of these then work them out left to right
    - Division includes **fractions**
  - **A**dditions or **S**ubtractions: + or -
    - If there are more than one of these then work them out left to right
- The acronym **BODMAS** or **BIDMAS** can help you remember the order of operations

In what order should fractions and roots be dealt with?

- **Fractions** mean **division** in calculations (BO**D**MAS/BI**D**MAS)
- There may be "invisible brackets" around the numerator and around the denominator
  - e.g. $\frac{2+5}{7-2}$ means $(2+5) \div (7-2)$
  - Instead of brackets we extend the fraction line to show exactly what is on the top and what is on the bottom
- **Roots** are p**O**wers (B**O**DMAS/B**I**DMAS)
- There may be "invisible brackets" under the root
  - e.g. $\sqrt{9+16}$ means $\sqrt{(9+16)}$
  - Instead of brackets we extend the top line on the root symbol to show everything that is to be rooted

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

How do I use a calculator for BIDMAS/BODMAS questions?

- Ensure that you enter a long or complicated calculation carefully
  - For most modern calculators, the calculation can be typed in exactly as it is written on paper
  - You may need to use additional brackets on older calculators

Your notes

Examiner Tips and Tricks

- Make sure that you always check that your answer seems sensible!

Worked Example

Work out $(5-3)+2\times7^2$.

Use **BODMAS**

First calculate anything inside "**B**"rackets, $5-3=2$, so the question becomes

$$2+2\times7^2$$

Then any p"**O**"wers, $7^2=49$

$$2+2\times49$$

followed by any "**M**"ultiplications and "**D**"ivisions, $2\times49=98$

$$2+98$$

and finally any "**A**"dditions and "**S**"ubtractions, $2+98=100$

$$\mathbf{(5-3)+2\times7^2=100}$$

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

Addition & Subtraction

How do I add large numbers without a calculator?

- There are a variety of written methods that can be used to add large numbers
  - The order in which numbers are added is **not** important
- The **column method** is the most commonly used hand written method
  - The numbers are written one number above the other,
    - Line up the digits using **place value columns**
  - Add each pair of corresponding digits from the top and bottom rows (work **right to left**)
  - If the result is a single digit
    - write the result in the relevant place value column below the line
  - If the result is a 2 digit number
    - the ones are written in the relevant place value column below the line
    - the tens are carried to the top of the next column
  - If the addition of the final pair of digits results in a 2 digit number
    - write both digits below the line
- For example, the addition 9789 + 563 = 10 352

$$\begin{array}{r} 1\,1\,1 \\ 9\,7\,8\,9 \\ +\quad 5\,6\,3 \\ \hline 1\,0\,3\,5\,2 \end{array}$$

How do I subtract large numbers?

- A variety of written methods exist, but you only need to know one
  - The order in which two numbers are subtracted **is** important so ensure the calculation is the right way round
- The **column method** is the most commonly used hand written method
  - The numbers are written one number above the other

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

    - Line up the digits using **place value columns**
    - The number being subtracted should be below the original amount
  - Subtract each digit in the bottom value from the corresponding digit in the top value (work **right to left**)
  - If the digit being subtracted is bigger than the one it is subtracted from
    - **"borrow ten"** from the next column to the left
- For example, 392 - 28 = 364

$$\begin{array}{rr} & 3\,\overset{8}{\cancel{9}}\,{}^{1}2 \\ - & 2\,8 \\ \hline & 3\,6\,4 \end{array}$$

What words are used for addition and subtraction?

- **Addition** may be phrased using the words: **plus, total** or **sum**
- **Subtraction** may be phrased using the words: **difference** or **take away**

Examiner Tips and Tricks

- A good way to check your answer without a calculator is to estimate it
  - e.g. if you work out 32 870 ÷ 865 to be 295, check by doing 30 000 ÷ 1 000 in your head which is 30, so your answer is probably wrong (the actual answer is 38)

Worked Example

(a) Find the sum of 3985 and 1273.

Notice that the word sum is used but this means add
Quickly estimate the answer

$$4000 + 1000 = 5000$$

Write the numbers in two rows and columns aligned

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

$$\begin{array}{r} 3985 \\ +1273 \\ \hline \end{array}$$

Start with the ones (units) column, writing the answer below the line but in the same column

$$\begin{array}{r} 3985 \\ +1273 \\ \hline 8 \end{array}$$

Move on to the tens (next on the left) column
The sum is 15 so the 5 (ones) is written below the line and the 1 (tens) 'carries over' to the next (hundreds) column

$$\begin{array}{r} 1 \\ 3985 \\ +1273 \\ \hline 58 \end{array}$$

Next is the hundreds column which again results in a two-digit answer

$$\begin{array}{r} 11 \\ 3985 \\ +1273 \\ \hline 258 \end{array}$$

Finally add the digits in the thousands column

$$\begin{array}{r} 11 \\ 3985 \\ +1273 \\ \hline 5258 \end{array}$$

Check the final answer is similar to your estimate; 5000 and 5258 are reasonably close

3985 + 1273 = **5258**

(b) Find the difference between 506 and 28.

Notice that the word difference is used; this means subtract
Quickly estimate the answer

500 - 30 = 470

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Write the numbers in two rows, column aligned, ensuring the top number is the number being subtracted *from*

$$\begin{array}{rrrr} & 5 & 0 & 6 \\ - & & 2 & 8 \\ \hline \end{array}$$

In the ones (units) column, 6 is smaller than 8, so borrow from the next column (tens)
The tens column is 0, so borrow from the column to the left of that (hundreds)

$$\begin{array}{rrrr} & 4 & & \\ & \cancel{5} & {}^{1}0 & 6 \\ - & & 2 & 8 \\ \hline \end{array}$$

This turns the 0 (in the tens column) into a 10 which we can then borrow from (for the ones column)

$$\begin{array}{rrrr} & 4 & 9 & \\ & \cancel{5} & \cancel{10} & {}^{1}6 \\ - & & 2 & 8 \\ \hline \end{array}$$

16 - 8 can be now be calculated in the ones column

$$\begin{array}{rrrr} & 4 & 9 & \\ & \cancel{5} & \cancel{10} & {}^{1}6 \\ - & & 2 & 8 \\ \hline & & & 8 \end{array}$$

Move onto the tens column which is 9 - 2 = 7
There is nothing to subtract in the last (hundreds) column (4 - 0 = 4)

$$\begin{array}{rrrr} & 4 & 9 & \\ & \cancel{5} & \cancel{10} & {}^{1}6 \\ - & & 2 & 8 \\ \hline & 4 & 7 & 8 \end{array}$$

Check the final answer is similar to the estimate; 470 and 478 are reasonably close

506 - 28 = **478**

Your notes

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

Multiplication

How do I multiply two numbers without a calculator?

- There are a variety of written methods that can be used to add large numbers
  - You only need to know **one** method, but be able to use it confidently
  - Four common methods are described below, but there are many other valid methods

How do I use the column method?

- This is an efficient method if you are confident with multiplication
- To use the **column method**:
  - Write one number above the other lining up the digits using **place value columns**
  - Multiply the first digit (on the right) from the bottom value by each digit in the top value
    - Write the result under the line with the digits in the correct place value columns
  - Multiply the next digit in the bottom value by each digit in the top value
    - Always work from **right to left**
    - Use 0s as place holders when multiplying digits in columns other than the ones column
  - For example, 87 × 426 = 37 062

$$\begin{array}{r} 8\ 7 \\ \times\ 4\ 2\ 6 \\ \hline 5\ 2\ 2 \\ 1\ 7\ 4\ 0 \\ 3\ 4\ 8\ 0\ 0 \\ \hline 3\ 7\ 0\ 6\ 2 \end{array}$$

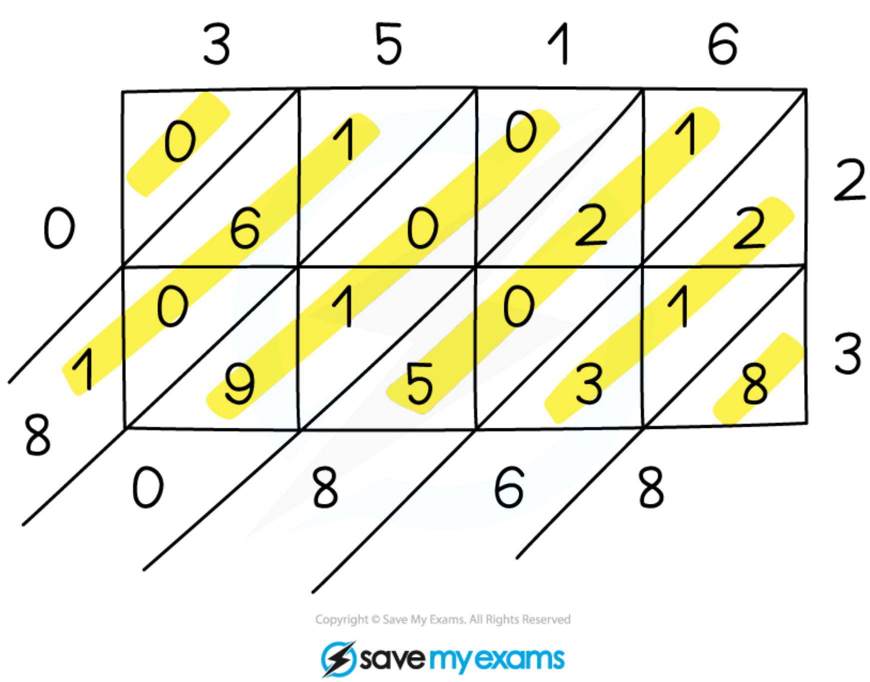
How do I use the lattice method?

- The lattice method is good for numbers with two or more digits
  - This method allows you to work with individual digits

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

- To use the **lattice method**:
  - Draw a grid
    - The **number of rows** should be the same as the number of digits in one number
    - The **number of columns** should be the same as the number of digits in the other number
    - Draw diagonal lines through the boxes
  - Multiply each pair of digits, writing the result in the relevant box
    - Ones should be written in the bottom half of the box and tens in the top half of the box
  - Add the digits along the diagonals and write the result in the diagonal outside the grid
    - Carry the tens of any 2 digit result into the next diagonal
- For example, 3516 × 23 = 80 868



How do I use the grid method?

- This method keeps the value of the larger number intact
  - It may take longer with two larger numbers
  - Be careful lining up numbers with lots of zeros!

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

- To use the **grid method**
  - Draw a grid
    - The **number of rows** should be the same as the number of digits in one number
    - The **number of columns** should be the same as the number of digits in the other number
  - Label the rows and columns with the values of each digit
    - E.g. For 3516 you would use 3000, 500, 10 and 6
  - Multiply together the relevant values and put the results in the boxes
  - Add up all of the cells in the boxes
- For example, 3516 × 7 = 24 612

	3000	500	10	6
7	21 000	3 500	70	42

Copyright © Save My Exams. All Rights Reserved
save my exams

$$
\begin{array}{r}
21\,000 \\
3\,500 \\
70 \\
+ \quad 42 \\
\hline
24612
\end{array}
$$

Copyright © Save My Exams. All Rights Reserved
save my exams

How do I use the repeated addition method?

- This is best for smaller, simpler cases

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

- You may have seen this called 'chunking'
- To use the **repeated addition method**
  - Build up to the answer using simple multiplication facts that can be worked out easily
  - To find 13×23 :
    $1 \times 23 = 23$

    $2 \times 23 = 46$

    $4 \times 23 = 92$

    $8 \times 23 = 184$
  - So, $13 \times 23 = 1 \times 23 + 4 \times 23 + 8 \times 23 = 23 + 92 + 184 = 299$

What words are used for multiplication and division?

- **Multiplication** may be phrased using the words **lots of, times** or **product**
- **Division** may be phrased using the words **quotient, share** and **per**

Examiner Tips and Tricks

- A good way to check your answer without a calculator is to estimate it (e.g. by rounding everything to 1 significant figure)

Worked Example

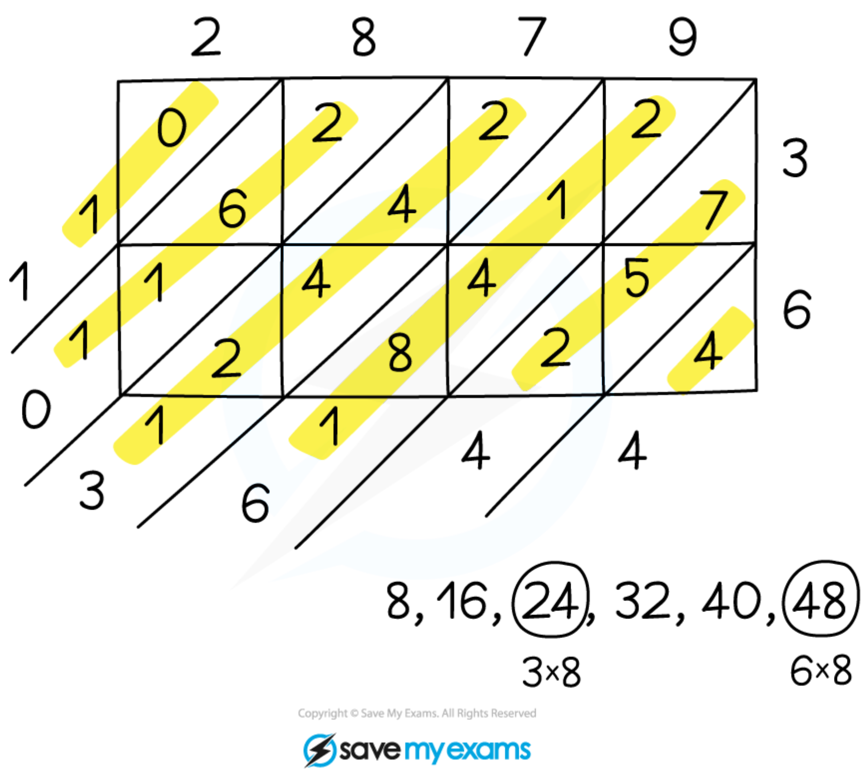
Multiply 2879 by 36.

As you have a 4-digit number multiplied by a 2-digit number then the lattice method is a good choice

Start with a 4×2 grid....

© 2015-2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes



Notice the use of listing the 8 times table underneath to help with some of the multiplication within the lattice

Use an estimate to check your answer; 3000×40 is equal to 120 000

2879 × 36 = **103 644**

Division

How do I divide a number by another without a calculator?

- The most common written method for division is short division (or "the bus stop method")
  - There are other methods such as long division, but short division is generally the most efficient
- While short division is best when dividing by a single digit, for bigger numbers you may need a different approach
- You can use other number skills to help

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

- eg. cancelling fractions, “shortcuts” for dividing by 2 and 10, and the repeated addition (“chunking”) method covered in Multiplication

Your notes

Short division (bus stop method)

- Unless you can use simple shortcuts such as dividing by 2 or by 10, this method is best used when dividing by a single digit
- To find 174 ÷ 3
  - Starting from the left; 3 fits into 1, 0 times, with a remainder of 1
    - Carry the remainder of 1 over to the next digit, which forms 17
    - $\begin{array}{r} 0\phantom{\,{}^{1}7\ 4} \\ 3\overline{)1\ {}^{1}7\ \ 4} \end{array}$
  - 3 fits into 17, 5 times, with a remainder of 2
    - Carry the remainder of 2 over to the next digit, which forms 24
    - $\begin{array}{r} 0\ \ 5\phantom{\ {}^{2}4} \\ 3\overline{)1\ {}^{1}7\ {}^{2}4} \end{array}$
  - 3 fits into 24, 8 times, with no remainder
    - $\begin{array}{r} 0\ \ 5\ \ \ 8 \\ 3\overline{)1\ {}^{1}7\ {}^{2}4} \end{array}$
  - So, 174 ÷ 3 = 58

Factoring & cancelling

- This involves treating division as you would if you were asked to simplify fractions
  - For example, 1008 ÷ 28 can be written as $\frac{1008}{28}$
  - This can then be simplified
    - $\frac{1008}{28} = \frac{504}{14} = \frac{252}{7}$
    - 252 ÷ 7 can then be calculated using short division; the answer is 36

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Dividing by 10, 100, 1000, ... (Powers of 10)

- This is a case of shifting digits along the place value columns

  For example

  - $380 \div 10 = 38.0$ (shifts by 1 column)
  - $45 \div 100 = 0.45$ (shifts by 2 columns)
  - $28 \div 1000 = 0.028$ (shifts by 3 columns)
    - For cases like this, it can help to add leading zeros
    - $0028 \div 1000$ may be easier to visualise

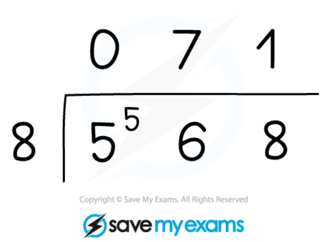
Your notes

Worked Example

Divide 568 by 8.

This is division by a single digit so short division would be an appropriate method
If you spot it though, 8 is also a power of 2 so you could just halve three times

Using short division, the bus stop method:



8 fits into 5, 0 times, with a remainder of 5
8 fits into 56, 7 times exactly
8 fits into 8, 1 time exactly

Use an estimate to check your answer; $600 \div 10$ is equal to 60

$568 \div 8 = \mathbf{71}$

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Operations with Decimals

Your notes

Operations with Decimals

How do I add or subtract decimals without a calculator?

- If the numbers involve **decimals**, the column methods for addition and subtraction can be used
  - Line the place value columns up carefully
  - Make sure the **decimal points** are **all in the same column**
    - Writing zeros (often called place value holders) can help keep everything in line
  - e.g. 2.145 + 13.02 would be written as $\begin{array}{r} 02.145 \\ +\ \ 13.020 \\ \hline . \end{array}$

Worked Example

Johnny has £32.50 and spends £1.74.

Calculate how much money Johnny has left.

This is a subtraction question as Johnny has *spent* money

Make a quick estimate

$$33 - 2 = 31$$

Align the digits by place value, ensuring £32.50 is the top number and using the decimal points as a starting point
Using place value holding zeros is optional

$$\begin{array}{r} 32.50 \\ -\ \ 01.74 \\ \hline . \end{array}$$

The column furthest right is the hundredths column
0 is smaller than 4 so borrow from the next (tenths) column

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

$$\begin{array}{r} 4 \\ 32.\cancel{5}^1 0 \\ -01.7\ 4 \\ \hline .\ \ 6 \end{array}$$

Next is the tenths column, but again 4 is smaller than 7 so borrow 10 from the ones column

$$\begin{array}{r} 1\ {}^1 4 \\ 3\cancel{2}.\cancel{5}^1 0 \\ -01.7\ 4 \\ \hline .\ \ 6 \end{array}$$

14 - 7 = 7 in the tenths column and continue working 'right to left'

$$\begin{array}{r} 1\ {}^1 4 \\ 3\cancel{2}.\cancel{5}^1 0 \\ -01.7\ 4 \\ \hline 30.7\ 6 \end{array}$$

Check the final answer is similar to the estimate
£31 and £30.76 are reasonably close

Johnny has **£30.76** left

How do I multiply decimals without a calculator?

- The standard methods for multiplication can be easily adapted for use with decimal numbers
  - E.g. column method, lattice method and grid method
- You can **make a problem easier** by converting to **integer values** then undoing the action to the answer
  - **Multiply** each value by a **power of 10** to make it an integer
  - Perform the easier multiplication
  - Undo the initial action by **dividing** by the **same powers of 10**
  - E.g. 2.5 x 4.01
    - Multiply 2.5 by 10 to give 25 and 4.01 by 100 to give 401
    - 25 x 401 = 10 025
    - Divide the answer by 10 and 100 to undo the initial changes

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

- So the answer is 10.025
- An alternative method is to **ignore** the **decimal point** whilst multiplying
  - then **put it back in** the correct place for the final answer using **estimation**
  - E.g. 1.3 × 2.3
    - Ignoring the decimals this is 13 × 23, which works out to 299
    - 1.3 × 2.3 is approximately 1.5 × 2, which is 3
    - So we know the answer should be close to 3, rather than 0.3 or 30
    - So the answer is 2.99

Your notes

Examiner Tips and Tricks

A good way to check your answer without a calculator is to estimate it by rounding everything to 1 or 2 significant figures.

Worked Example

Without using a calculator, find 1.57 × 0.78 .

Method 1

Multiply both values by powers of 10 to create a calculation using integer values

$$1.57 \times 100 = 157$$
$$0.78 \times 100 = 78$$

$$157 \times 78$$

Using the grid method

	100	**50**	**7**
70	7000	3500	490
8	800	400	56

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

$$\begin{array}{r} 7000 \\ +3500 \\ 490 \\ 800 \\ 400 \\ 56 \\ \hline 12246 \end{array}$$

157 × 78 = 12 246

Undo the initial action by dividing the answer by 10 000
(This is dividing by 100 and then 100 again)

12 246 ÷ 10 000

1.2246

Method 2

Ignore decimal point to form a different, non-decimal calculation

157 × 78

Using the grid method

	100	50	7
70	7000	3500	490
8	800	400	56

$$\begin{array}{r} 7000 \\ +3500 \\ 490 \\ 800 \\ 400 \\ 56 \\ \hline 12246 \end{array}$$

157 × 78 = 12 246

Estimate the original calculation to write this as the correct answer

1.57 × 0.78 is approximately 2 × 1 = 2

So the correct answer is between 1 and 10

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

1.2246

Your notes

How do I divide a decimal without a calculator?

- Use a similar approach to when multiplying decimals
- **Make a problem easier** by converting to **integer values** then undoing the action to the answer
  - Write the division as a **fraction**
  - **Multiply** each value by a **power of 10** to make it an integer
  - Perform the easier division
  - Undo the initial action
    - **divide** by the **same power of 10** the **numerator** was multiplied by
    - **multiply** by the **same power of 10** the **denominator** was multiplied by
  - E.g. 37.5 ÷ 0.25
    - $37.5 \div 0.25 \rightarrow \frac{37.5}{0.25}$
    - $\frac{37.5 \times 10}{0.25 \times 100} = \frac{375}{25}$
    - $\frac{375}{25} = 15$
    - Undo the initial actions $\frac{15 \times 100}{10}$
    - So the answer is 150
- Alternatively, ignore the decimal point whilst carrying out the division
  - then **put it back in** the correct place for the final answer using **estimation**
  - E.g. 4.68 ÷ 6
    - 4.68 ÷ 6 is approximately 5 ÷ 6, which is between 1 and 0.5
    - So we know the answer should be 0.78, rather than 7.8 or 0.078

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

Your notes

Worked Example

Without using a calculator, find $69.02 \div 0.7$.

Method 1

Write as a fraction

$$\frac{69.02}{0.7}$$

Multiply both numbers by powers of 10 to create a calculation with integer values

$$\frac{69.02 \times 100}{0.7 \times 10} = \frac{6902}{7}$$

Use short division (bus stop method)

$$7\overline{)6\,{}^{6}9\,{}^{6}0\,{}^{4}2}\quad \text{quotient: } 0\ 9\ 8\ 6$$

$$\frac{6902}{7} = 986$$

Convert the answer
Divide by 100 to cancel out 69.02 being multiplied by 100
Multiply by 10 to cancel out 0.7 being multiplied by 10

$$\frac{986 \times 10}{100}$$

98.6

Method 2

Ignore the decimal point to form a different, non-decimal calculation

$$6902 \div 7$$

Use short division (bus stop method)

$$7\overline{)6\,{}^{6}9\,{}^{6}0\,{}^{4}2}\quad \text{quotient: } 0\ 9\ 8\ 6$$

$$6902 \div 7 = 986$$

Estimate the original calculation to write this as the correct answer

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers

$$69.02 \div 0.7 \text{ is approximately } 70 \div 1 = 70$$

So the correct answer is between 10 and 100

98.6

Your notes

© 2015–2025 Save My Exams, Ltd. · Revision Notes, Topic Questions, Past Papers